

Working Example: From Black-Box Re-ranking to Semantic Explanations

To illustrate how our framework operates, we provide a simplified scenario using a restaurant recommendation dataset (e.g., Yelp).

The Setup:

- **User (u):** Alice.
- **Target Protected Group (Gprot):** "Local Cuisine" (The fairness objective is to increase the visibility of local businesses).
- **List Size (K):** 4 items.

Step 1: Top-N Recommendation & Fairness Re-ranking

First, the baseline recommender model M generates a list for Alice based purely on her historical preferences (mostly Japanese and Fast Food). Then, the fairness-oriented re-ranker R modifies this list to meet the fairness metric (e.g., FSym), boosting "Local Cuisine" items.

Rank (pos)	Baseline List (Lubase)	Attributes (xi)	Re-ranked List (Lure)	Attributes (xi)
1	Sushi Palace	Japanese, Sushi	Casa Tapas	Local Cuisine
2	Burger Town	Fast Food, Burger	Olive Grove	Mediterranean
3	Olive Grove	Mediterranean	Sushi Palace	Japanese, Sushi
4	Casa Tapas	Local Cuisine	Burger Town	Fast Food, Burger

Step 2: Attribute Level Outcome Analysis

Now, the framework operates locally on these lists. For each item in the baseline, we compute the position delta $\Delta(i,u)$ and assign a binary label $y(i,u)$ representing promotion (+1) or demotion (-1).

- **Item: "Olive Grove"** (Attributes: Mediterranean)
 - Baseline pos: 3 | Re-ranked pos: 2
 - $\Delta(i,u)=2-3=-1$
 - Since $\Delta < 0 \rightarrow y=+1$ (**Promotion**)
- **Item: "Sushi Palace"** (Attributes: Japanese, Sushi)
 - Baseline pos: 1 | Re-ranked pos: 3
 - $\Delta(i,u)=3-1=+2$
 - Since $\Delta > 0 \rightarrow y=-1$ (**Demotion**)

Step 3: Explainable Rule Induction

We construct our training dataset $D=\{(x_i,y(i,u))\}$ from these outcomes across all users. The surrogate decision tree extracts patterns mapping item attributes (x_S) to the re-ranking decision (y).

In this scenario, the tree induction discovers the following rules (Crule):

- **Rule A:** IF Tag == "Mediterranean" THEN $y = +1$ (*Promotion Rule*)
- **Rule B:** IF Tag == "Japanese" THEN $y = -1$ (*Demotion Rule*)

Step 4: Explanation Generation

Finally, we map the extracted conditions (Crule) and the fairness target (Gprot = "Local Cuisine") into our predefined natural language templates:

- Case 1: Side-effect Promotion (Applying Rule A to "Olive Grove")

*"Focusing on **Local Cuisine** had the side effect of promoting **Mediterranean** restaurants as a similar alternative to your usual spots."*

- Case 2: Granular Demotion (Applying Rule B to "Sushi Palace")

*"To boost **Local Cuisine**, we specifically lowered the rank of **Japanese** restaurants from your recommendations."*